

The Infrastructure–Isolation Principle

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Reliance, semantic neighborhoods, and proof complexity in Mathlib

A formal library grows by reuse. Definitions support lemmas, lemmas support later theorems, and automation hides long chains of accumulated infrastructure behind short tactic scripts. This suggests two competing pictures of maturity. Reuse could make deep results resemble one another: shared machinery might concentrate the library into dense families. Or reuse could enable specialization: common infrastructure might support increasingly particular endpoints whose proofs have few close analogues.

Measurements on 10,000 theorem–proof pairs from LeanDojo Benchmark 4 support the second picture [4]. Theorems resting on more kernel-visible infrastructure have more involved recorded proofs, but smaller semantic neighborhoods. The contraction is stronger for proofs than for statements. We call this joint pattern the *Infrastructure–Isolation Principle*:

In a mature formal library, declarations that depend on more mathematical infrastructure tend to have more involved proofs but fewer close semantic peers; procedural specialization outpaces semantic specialization.

Intuitive interpretation

The dependency graph measures how much mathematical scaffolding a theorem stands on. The embeddings measure how many close companions it has, and the complexity score measures how involved its recorded construction is. The surprise is that more scaffolding accompanies fewer companions—especially for proofs. This is an empirical pattern, not a theorem of Lean.

Reliance in the kernel graph. Let $G = (V, E)$ be the directed graph of declarations in the pinned Mathlib environment. An edge $u \rightarrow v$ means that declaration v uses constant u in its kernel-checked type or value, including the elaborated proof term. The export contains 888,992 declarations and 13,264,773 direct edges. Unlike identifiers written visibly in tactic syntax, these edges include constants introduced by automation such as `simp`.

This graph-first view is also entering learned proving. Graph2Tac builds hierarchical representations of new formal concepts from the dependency structure of a Coq development and updates them online as the library grows [5]. Here the graph becomes an object of measurement in its own right: not only a route to the next tactic, but a record of how accumulated infrastructure positions a theorem.

For target declaration i , let $A(i)$ contain every upstream declaration that can reach i , and let $d(u, i)$ be the length of the shortest directed path from u to i . With discount $0 < \alpha \leq 1$, define dependency mass and its reported reliance score by

$$D_\alpha(i) = \sum_{u \in A(i)} \alpha^{d(u, i)}, \quad R_i = \log(1 + D_{0.5}(i)).$$

Here $\log$ is the natural logarithm. At $\alpha = 1$, dependency mass counts distinct ancestors; $\alpha = 0.5$ emphasizes nearby infrastructure without discarding the transitive closure. All 10,000 searches terminate within 52 levels. The median target has 39 direct dependencies, shortest-path depth 18, and $R_i = 4.668$.

Connectedness in two semantic spaces. The same Cohere Embed v4 map used in the earlier semantic study sends statement text and proof text separately to vectors $e_i^{(S)}, e_i^{(P)} \in \mathbb{R}^{1024}$ [2]. For view $v \in \{S, P\}$, normalize

$$x_i^{(v)} = \frac{e_i^{(v)}}{\|e_i^{(v)}\|_2}, \quad \|e\|_2 = \left(\sum_k e_k^2 \right)^{1/2}.$$

The dot product $x_i^{(v)} \cdot x_j^{(v)}$ is then cosine similarity. Let $\mathcal{N}_{100}^{(v)}(i)$ be the exact top-100 neighbors of i in view v . A view-specific threshold τ_v is the 99th percentile of 500,000 fixed random-pair cosines: $\tau_S = 0.690$ and $\tau_P = 0.755$. For $j \in \mathcal{N}_{100}^{(v)}(i)$, set

$$w_{ij}^{(v)} = \max\{0, x_i^{(v)} \cdot x_j^{(v)} - \tau_v\}, \quad p_{ij}^{(v)} = \frac{w_{ij}^{(v)}}{\sum_{\ell \in \mathcal{N}_{100}^{(v)}(i)} w_{i\ell}^{(v)}}.$$

The effective semantic neighborhood is

$$S_i^{(v)} = \exp\left(-\sum_{j \in \mathcal{N}_{100}^{(v)}(i)} p_{ij}^{(v)} \log p_{ij}^{(v)}\right),$$

with $S_i^{(v)} = 0$ if every weight is zero. This is the entropy-equivalent number of neighbors sharing the retained similarity weight: one dominant neighbor gives a value near one, and many balanced neighbors give a larger value, capped here at 100. Median connectedness is 31.3 in statement space but only 8.6 in proof space.

Complexity of the recorded proof. Let s_i be tactic-step count, r_i direct kernel-reference count, h_i maximum dependency depth, and q_{it} the fraction of proof steps whose tactic head is t . Tactic-head entropy is $H_i = -\sum_t q_{it} \log q_{it}$. If $z(y) = (y - \bar{y})/\text{sd}(y)$ standardizes a component over the 10,000 targets, define

$$C_i = \frac{1}{4} [z(\log(1 + s_i)) + z(\log(1 + r_i)) + z(h_i) + z(H_i)].$$

Thus C_i is an equal-weight index of observed structural involvement. It is not proof length alone, and it is not a claim about intrinsic mathematical difficulty.

The empirical sign pattern. Write $\rho(X, Y)$ for Spearman rank correlation, the ordinary Pearson correlation of the ranks of X and Y . The central observation is

$$\rho(R, C) = 0.734 > 0, \quad \rho(R, S_P) < \rho(R, S_S) < 0, \quad \rho(C, S_P) < \rho(C, S_S) < 0,$$

with measured values

$$\rho(R, S_S) = -0.314, \quad \rho(R, S_P) = -0.431, \quad \rho(C, S_S) = -0.398, \quad \rho(C, S_P) = -0.584.$$

Statement and proof connectedness remain positively but incompletely aligned: $\rho(S_S, S_P) = 0.490$. The pattern is therefore not “everything difficult is far from everything else.” It says that structural accumulation and semantic density point in opposing directions, while statement and proof geometries preserve a partial bridge.

The positive R – C association is partly built into the definitions: C contains direct reference count and dependency depth. The negative connectedness associations are not. Neither S_S nor S_P uses the dependency graph or tactic counts. They supply the nontrivial half of the principle.

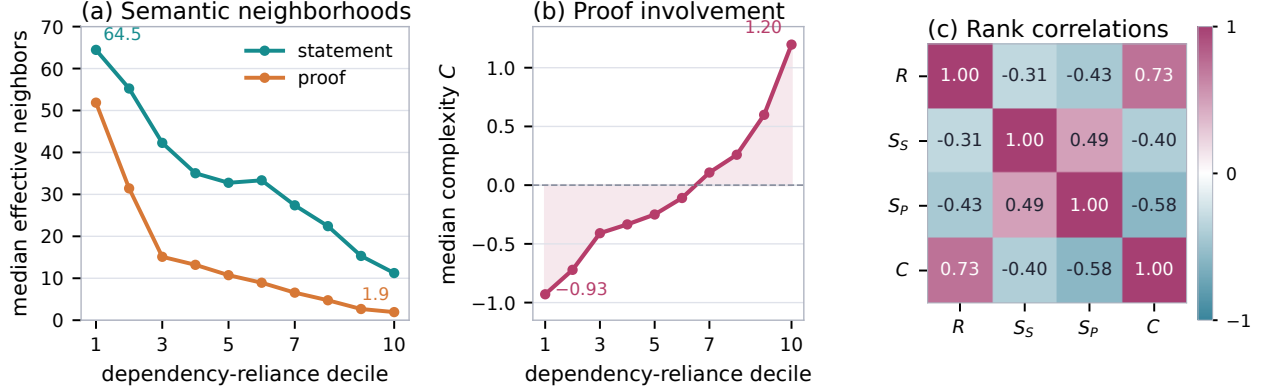


Figure 1: The Infrastructure-Isolation Principle in the frozen 10,000-target sample. Equal-count reliance deciles show (a) shrinking statement and proof neighborhoods and (b) increasing proof complexity. Panel (c) gives Spearman rank correlations among reliance R , statement connectedness S_S , proof connectedness S_P , and complexity C . Effective neighborhood medians, rather than embedding projections, are shown.

Proposition (the infrastructure rank-sign law). Let $U_X = \Pr(X' < X) + \frac{1}{2} \Pr(X' = X)$ be the population midrank of $X \in \{R, C, S_S, S_P\}$. Suppose a latent infrastructure coordinate Z makes $m_X(z) = \mathbb{E}[U_X | Z = z]$ nondecreasing for $X = R, C$ and nonincreasing for $X = S_S, S_P$. If the rank residuals are pairwise conditionally uncorrelated given Z , then

$$\rho(R, C), \rho(S_S, S_P) \geq 0, \quad \rho(A, B) \leq 0 \quad \text{for } A \in \{R, C\}, B \in \{S_S, S_P\}.$$

Proof. Total covariance reduces the left side below to the right; for an independent Z' ,

$$2 \text{Cov}(U_X, U_Y) = \mathbb{E}[(f(Z) - f(Z'))(g(Z) - g(Z'))].$$

Here $f = m_X$ and $g = m_Y$. The product has the sign claimed when the functions move in the same or opposite directions; rank standardization preserves it. Strict sign on a positive-probability set gives strict inequality. $\square$

Intuitive interpretation

Imagine moving along one hidden “infrastructure” dial. Reliance and complexity rise as the dial turns, while both kinds of neighborhood shrink. That alone forces the observed correlation directions. It does not determine how strong the correlations should be, and the faster isolation of proofs remains empirical.

Corollary (the one-factor tetrad test). Specialize further to standardized rank variables

$$\tilde{U}_R = a_R Z + \varepsilon_R, \quad \tilde{U}_C = a_C Z + \varepsilon_C, \quad \tilde{U}_{S_S} = -a_S Z + \varepsilon_S, \quad \tilde{U}_{S_P} = -a_P Z + \varepsilon_P,$$

where $a_R, a_C, a_S, a_P > 0$, $\text{Var}(Z) = 1$, and the residuals are mutually uncorrelated and uncorrelated with Z . Pairwise correlations are then signed products of loadings, so

$$\rho(R, C)\rho(S_S, S_P) = \rho(R, S_S)\rho(C, S_P) = \rho(R, S_P)\rho(C, S_S).$$

Intuitive interpretation

If one linear hidden dial controlled everything, the three correlation products would agree. They do not: the measured values are 0.360, 0.183, and 0.172. One dial therefore explains the directions but not their strengths. Nonlinearity, correlated residual structure, or an additional procedural axis is needed to explain the remaining geometry.

A gradient, not only a correlation. Figure 1 orders targets by R_i and divides them into ten equal-count groups. From lowest to highest decile, median statement neighborhood falls from 64.5 to 11.2, proof neighborhood from 51.9 to 1.9, and complexity rises from -0.93 to 1.20 . Proof connectedness decreases at every step; statement connectedness has one small reversal. Rankings remain stable: tested dependency discounts correlate at least 0.904 , semantic thresholds at least 0.926 , and leave-one-component-out complexity scores at least 0.941 with the primary rankings.

Why proofs isolate faster than statements. Statements belong to recurring mathematical families; recorded tactic proofs are implementations shaped by available lemmas, automation, notation, and intermediate goals. Common infrastructure can support similar claims without enforcing similar proof artifacts. Earlier feature topics and embeddings separated content from procedure while retaining a local bridge [1, 2]. The present result adds direction from the kernel graph: increasing dependency depth weakens both neighborhoods, but weakens the procedural one more. Lean Copilot already places tactic suggestion, proof search, and premise selection behind one Lean-native learned interface [6]; our measurements suggest that such systems may benefit from treating semantic family and procedural precedent as related but distinct retrieval signals.

A two-axis map. Reliance and connectedness distinguish reusable basics, unusual elementary results, mature specialized families, and bespoke endpoints atop deep infrastructure. The last is a retrieval frontier: semantic examples can help generation [3], but a high-reliance, low-connectedness theorem needs broad library concepts while offering few procedural precedents. Retrieval for this tail should combine semantic proximity with dependency coverage. LeanSearch v2 sharpens the same frontier as *global premise retrieval*: recovering a scattered supporting set rather than one nearest declaration [7].

Scope of the principle. These scores describe one snapshot, one recorded proof per theorem, and one encoder. Dependency mass includes formalization machinery; scripts are not canonical; neighborhoods are model-dependent; and the analysis is observational. Replication across snapshots, libraries, and multiple proofs should test the principle. Within those limits, the result joins structural and learned geometry: common foundations support specialized conclusions whose proof artifacts become distinctive faster than their statements do. Accumulated infrastructure produces specialization, not homogenization.

References

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